

Caio Jordão de Lima Carvalho

SOFTWARE DEVELOPER

Estrada das Barreiras, 460, BL 186B AP 101, Cabula, Salvador, Bahia, Brazil

(+55) 71 99703-3224 | caiojcarvalho@gmail.com | caiojcarvalho.wordpress.com | cjlcarvalho | cjlcarvalho

Summary

Current Artificial Intelligence Researcher at Supercomputing Center for Industrial Innovation - SENAI CIMATEC. 3+ years experience specializing in the back-end/data engines development, information retrieval, and machine learning. Super nerd, easy learner, Linux lover and KDE contributor. Problem solver, interested in new challenges, and developing new tools.

Work Experience

Supercomputing Center for Industrial Innovation - SENAI CIMATEC

Salvador, Bahia, Brazil

ARTIFICIAL INTELLIGENCE RESEARCHER

Dec. 2019 - Present

- Worked directly on a project involving machine learning methods for predictive maintenance.
- Implemented and evaluated clustering approaches (KNN, RRCF, Bootstrap, Luminol) for anomaly detection.
- Built pipelines focused on exploratory data analysis, including data cleaning, filtering, normalization and statistical evaluation.
- Implemented deep learning methods (LSTM, GRU, MLP, CNN) for time-series prediction.

Escavador

Salvador, Bahia, Brazil

BACK-END DEVELOPER

Oct. 2018 - Nov. 2019

- Designed web crawler architectures for systems focused on information retrieval of Brazilian legal cases and laws, which extracts millions of documents from different courts from Brazil, including the complete legislation from the Brazilian territory (<https://www.escavador.com/jurisprudencia>, <https://www.escavador.com/legislacao>).
- Implemented several API microservices using Python, asynchronous libraries and MySQL.
- Project team management.
- Built scripts to extract juridical information from many governmental gazettes.
- Implemented artificial intelligence models focused on information filtering and natural language processing.
- Integrated search engines with support for machine learning ranking using vespa.ai.
- Monitored several web crawlers using Grafana and Prometheus.
- Built fully automated CI pipelines on Gitlab CI for containerized applications using Docker.

KDE Community

Remote

SOFTWARE DEVELOPER & CONTRIBUTOR

Dec. 2017 - Present

- KDE e.V. member.
- Improved SMART support and KAuth operation flow in KDE Partition Manager during Season of KDE 2018.
- Implemented LVM Volume Group support in Calamares and MDRAID support in KDE Partition Manager during Google Summer of Code 2018.
- Mentored students during Google Code-in 2018, helping them to complete tasks for KDE Partition Manager.
- Mentor representing KDE during Google Code-in 2018 Summit in Google HQ (San Francisco, CA).
- Mentor during Google Summer of Code 2019 in two projects: "Improving Khipu and Analitza" and "Port Authentication to Polkit-qt".
- My patches to KDE Partition Manager and Calamares helped me to improve my skills with C++, C, Qt (widgets), CMake, SMART (libatasmart, smartctl), Dbus, mdadm, dmraid, lvm and Linux partitioning tools.

Labrasoft - IFBA

Salvador, Bahia, Brazil

UNDERGRADUATE RESEARCH ASSISTANT

Aug. 2017 - Aug. 2019

- Developed a simulated stock market to evaluate bots behaviour in decision-making processes using MQL5, MetaTrader and Python.
- Deployed object tracking algorithms based on Deep Learning approaches in embedded systems using Intel Movidius, Raspberry Pi and Python.
- Implemented a generative-adversarial model for image enhancement during Minimally Invasive Surgeries.

TecnoTRENDS

Salvador, Bahia, Brazil

SOFTWARE DEVELOPMENT INTERN

Oct. 2016 - Oct. 2017

- Developer of a module for an educational ERP system focused on elementary schools using C#, ASP.NET and SQL Server.
- Released versions from this module for all the elementary schools from the city of Feira de Santana, Bahia.
- Development of new features, graphical interfaces and SQL routines.

Honors & Awards

2018	Grand Prize Winner , Season of KDE 2018	Remote
2018	Winner , Innovative Challenge - IFBA Projects' Hotel	Salvador, Brazil
2017	Gold Medalist , ProgBASE 2017 - Competitive Programming Contest	C. das Almas, Brazil
2018	Bronze Medalist , 2nd Programming Contest of Bahia	Salvador, Brazil

Relevant Presentations

Akademy 2019

PRESENTER FOR <KPMCORE IMPLEMENTATIONS AND GOALS FOR THE FUTURE>

- Talked about my work on kpmcore and described some updates and features from KDE Partition Manager 4.0.

Milan, Italy

September, 2019

QtCon Brazil 2018

PRESENTER FOR <COMPUTER VISION WITH QT QUICK AND OPENCV>

- Introduced how to integrate the OpenCV library with Qt Quick in mobile environments.

São Paulo, Brazil

Nov. 2018

V Workshop Labrasoft-GPETEC

PRESENTER FOR <INTRODUCTION TO KERAS AND DEEP LEARNING MODELS>

- Explained about the Keras library, including an introduction to deep learning models, such as Convolutional Neural Networks, Autoencoders and Generative Adversarial Networks.

Salvador, Brazil

Sep. 2018

ERBASE - Encontro Regional de Computação Bahia-Alagoas-Sergipe

PRESENTER FOR <ALPHA3 - EXPERT ADVISOR USING ARTIFICIAL NEURAL NETWORKS AND GENETIC ALGORITHMS>

- Explained about the Alpha3 project, which is an expert advisor to predict stock market trends based on neural networks and genetic algorithms.

Aracaju, Brazil

Aug. 2018

Publications

Alpha3 - Expert Advisor using Neural Networks and Genetic Algorithms to Predict Stock Market Trends

ERBASE 2018

CO-AUTHOR

Aug. 2018

- Paper published in ERBASE 2018. Explained about Alpha3 and its results on the prediction of the Brazilian stock market.

Organizer

2018 **Organizer**, WebMedia 2018

Salvador, Brazil

2018 **Organizer**, ADS-Integra & ADS-Gênero

Salvador, Brazil

2017 **Problem Writer**, IFBA Competitive Programming Contest

Salvador, Brazil

Education

IFBA (Federal Institute of Science, Education and Technology of Bahia)

Salvador, Brazil

B.TECH. IN SYSTEMS ANALYSIS AND DEVELOPMENT

Jun. 2016 - Oct. 2019

- GPA: 8.6 / 10.0
- Graduation Thesis Title: "Deep Autoencoder Approach to Enhance Low-Light Images during Minimally Invasive Procedures".
- Relevant Coursework: Data Structures, Algorithms, Software Architecture, Design Patterns, Distributed Systems, Software Engineering, Object-Oriented Design, Software Project Management.
- Technical Skills: C++ (3yrs), C, Python, Java, Git, Django, Qt, SQL.
- I was the student representative of this program while studying at IFBA.

College of the Rockies

Cranbrook, Canada

EXCHANGE STUDENT IN APPLIED SCIENCES

Jan. 2019 - May. 2019

- GPA: A-
- Relevant Coursework: Statistics, Probability, Mathematics.
- Selected for a scholarship to the College of the Rockies by the Emerging Leaders in the Americas Program.

Extracurricular Activity

ProgIFBA - Competitive Programming Group at IFBA

Salvador, Brazil

MEMBER

Dec. 2016 - Jan. 2019

- Created a group focused on competitive programming studies at my university.
- Proposed some training sessions and participated in some contests that have happened in my city.

Sonata Project

Salvador, Brazil

MEMBER

Feb. 2018 - Apr. 2019

- Development of a mobile platform to help people on learning how to read music scores.
- Studied about music information retrieval.
- Worked with a team to submit this project to a innovative challenge at IFBA. Our project was the winner in this competition.